Project Planning & Management

Project Name: Global Energy Shock 2026: Oil, Trade & Economic Exposure Analysis

Program: Digital Egypt Pioneers Initiative (DEPI), Data Analytics Track

1. Project Proposal

1.1 Project Name & Core Concept

This project builds a **decision-support platform** that links global oil price shocks to country-level trade balances, exchange rates and GDP. It brings eight data files into one database and one Power BI dashboard, so a user can pick a crude price scenario and immediately see which countries pay more, which countries earn more, and by how much.

The platform covers **150 countries**. It answers one practical question: *if crude oil gets more expensive, who is hit hardest, and how big is the hit compared with the size of their economy?*

1.2 The Real-World Problem

The data needed to answer this question exists, but it sits in different places:

- **Different agencies.** Production and prices come from the U.S. Energy Information Administration (EIA). GDP and macro indicators come from the World Bank. Crude trade comes from UN Comtrade. Oil consumption comes from Our World in Data / Energy Institute.
- **Different frequencies.** Production and Brent/WTI prices are **monthly** (Jan 2010 to May 2026). GDP, trade, consumption and exchange rates are **annual** (2010 to 2025).
- **Different units and definitions.** Trade is in USD, consumption is in TWh, and exchange rates are in local currency per USD. Imports and exports are also valued on different bases (CIF-type for imports, FOB-type for exports).

Because of this, decision-makers cannot quickly calculate how a crude price spike changes a country's **local-currency import bill** or how large that bill is compared with its GDP. Analysts end up joining files by hand, and the results are hard to repeat or check.

1.3 Project Objectives

1. **Integrate 38,922 rows across 8 datasets** into one ISO3 star-schema database. Success means no duplicate keys, no orphan country codes, and every table linked to the country table.
2. **Classify the 150 countries as Net Importers or Net Exporters** using net crude trade (imports minus exports).
3. **Calculate exposure ratios:** Crude Import Burden (% of GDP), Crude Export Dependence (% of GDP) and Oil Intensity (TWh per $ billion GDP). Each ratio is checked in SQL and again in Power BI.
4. **Build an interactive Power BI dashboard** with What-If sliders for +10%, +25% and +50% price shocks, to show the immediate financial exposure of each country in USD and local currency.

1.4 What the Project Will Not Do

The scenarios are simple financial calculations on observed trade values. They keep traded volumes fixed.

- It does not forecast future oil prices.
- It does not claim that any one event caused a price movement or a GDP loss.
- It does not model how countries would change their buying, selling or policy in response to a shock.

1.5 Dataset Overview & Source Provenance

The eight CSV files come from a Kaggle dataset titled *Global Energy Shock 2026: Oil, Trade & Economic Exposure*. The dataset combines publicly available data from the institutions listed above. The original publishers keep the rights to the underlying data, so the final submission must cite them and follow their usage terms.

#	Table	What It Holds	Coverage	Rows
1	`countries`	ISO codes, country name, region, income group, capital, latitude, longitude	150 countries	150
2	`monthly_oil_production`	Monthly total liquids and crude oil production	Jan 2010 to May 2026	28,959
3	`Monthly_oil_prices`	Monthly Brent and WTI prices	Jan 2010 to May 2026	197
4	`Macro_indicators_annual`	GDP, GDP per capita, GDP growth, inflation, trade % of GDP	2010 to 2025	2,372
5	`Oil_trade_annual`	Annual crude imports and exports (HS 2709)	2010 to 2025	2,400
6	`Oil_consumption_annual`	Annual oil consumption in TWh	2010 to 2025	2,294
7	`Exchange_rates_annual`	Official exchange rate (local currency per USD)	2010 to 2025	2,400
8	`Country_exposure_scores`	One row per country with net trade, importer/exporter flag, import burden, export dependence, oil intensity, local-currency import bill	Snapshot, no year column	150

How the tables connect. `countries` is the center of the star schema. The tables with a country code join to it on **ISO3**. Annual tables also join on **year** and monthly tables on **month**. Prices have no country code because they are global. File 08 has no year column, so its reference year(s) will be documented in Phase 2 before its numbers are compared with other tables.

V2 methodology: Missing is not zero. The dataset keeps missing source values as they are. A missing export value does not mean zero exports. We follow the same rule: a ratio is left blank when any input is missing, no value is filled in, and every chart shows how many countries it covers.

Important: The V2 dataset has no price-shock scenarios. The +10%, +25% and +50% scenarios are built by our team on top of the observed data.

1.6 Tech Stack

Tool	How We Use It
Excel	First look at the data, sampling and quality checks
SQL	Database schema, joins, aggregations, year-over-year changes and rankings with window functions
Power BI	Data model, DAX measures, What-If sliders and interactive maps
GitHub	Version control and shared project files

2. Project Plan & Timeline

2.1 Phases

Phase	Name	Dates (2026)	Main Work	Deliverable
1	Initiation & Requirements	Oct 10 to Oct 16	Scope, KPIs, roles, repository setup, Phase 1 documents	Phase 1 Documentation
2	Data Profiling & Audit	Oct 17 to Oct 23	Check all 8 files for missing values, duplicate keys and valid ISO3 codes; document file 08	Data Audit Log
3	SQL Database Design & ETL Pipeline	Oct 24 to Oct 31	Build the star schema, date and year tables, load clean data	SQL Schema & Clean Datasets
4	Exploratory Analysis & Metrics Calculation	Nov 01 to Nov 08	Importer/exporter classification, six KPIs, rankings	SQL Queries & Analysis Summary
5	Power BI Dashboard & Scenario Simulator	Nov 09 to Nov 18	Data model, DAX, What-If sliders, maps, drill-through	Power BI file (.pbix)
6	Quality Assurance & Metric Validation	Nov 19 to Nov 24	Compare SQL and Power BI results, test missing-value handling, user testing	Test Cases & Validation Report
7	Final Documentation, Video Demo & GitHub Release	Nov 25 to Nov 30	Final documents, demo video, slides, GitHub release	Final Submission Package

2.2 Gantt Chart

Each column is one calendar week. Phase dates may start or end inside a column.

Phase	Oct 10-16	Oct 17-23	Oct 24-30	Oct 31-Nov 6	Nov 7-13	Nov 14-20	Nov 21-27	Nov 28-30
1. Initiation & Requirements	■■■							
2. Data Profiling & Audit		■■■						
3. SQL Design			■■■	■				
4. Analysis & Metrics				■■	■			
5. Dashboard & Simulator					■■	■■■		
6. QA & Validation						■	■■	
7. Final Delivery							■	■■

2.3 Milestones

- **Oct 16:** Phase 1 documents approved
- **Oct 31:** Database loaded with zero orphan keys
- **Nov 08:** All six KPIs calculated in SQL
- **Nov 18:** Dashboard with working What-If sliders
- **Nov 24:** SQL and Power BI results match
- **Nov 30:** Final submission

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3. Task Assignment & Roles

Member	Role	Responsibilities
[Kareem Hassan Ahmad]	Project Leader & Business Analyst	Project coordination, progress tracking, analysis questions, and deliverables
[Kareem Hassan Ahmad + Mahmoud Essam Salah]	Data Cleaning Analyst	Profiling and cleaning all eight CSV files; documenting transformations
[Mahmoud Essam Salah]	SQL & Data Modeling Analyst	Creating relationships and queries; verifying keys

Member	Role	Responsibilities
[Abdelrahman Sherif Mohammad]	Historical & Economic Analyst	Analyzing oil prices, Iranian production, trade, and economic indicators
[Manar Farag Ahmad + Shrouk Mesbah Elsayed Mahdy]	Power BI & DAX Developer	Building dashboard pages, measures, filters, and visuals
[Ahmed Abdelrahman Mohammad]	Scenario Analysis & QA Analyst	Testing hypothetical shocks, reviewing calculations, and documenting limitations

Shared responsibility: Every member should understand the full data model, review the results and help prepare the final presentation. The Project Lead holds a progress check every two weeks.

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4. Risk Assessment & Mitigation Plan

#	Risk	What Could Go Wrong	Mitigation
1	**Missing values in trade and consumption data**	Treating a gap as zero makes a country look like it has no oil trade, and global totals and rankings become wrong.	Follow the V2 rule strictly: never fill missing values with zero. Calculate only where all inputs exist, and show the number of countries behind every result. Compare only complete sub-samples.
2	**Monthly prices vs. annual macro data**	Monthly Brent/WTI prices cannot join directly to annual GDP or trade. A wrong join duplicates rows and inflates results.	Build a date table and a year table. Average monthly prices to annual values in SQL. Check row counts after every join.
3	**Exchange-rate changes and CIF/FOB differences**	A weak or volatile currency changes the local-currency bill. Imports and exports are also valued on different bases, which can distort net trade.	Use only official annual rates (LCU per USD). Label local-currency results as annual estimates. State the CIF/FOB difference wherever net trade is shown.
4	**Team dependency bottlenecks**	If the SQL work is late, the Power BI work cannot start, and the dashboard has only 10 days.	Prototype Power BI early with sample data in the same structure. Work across roles. Hold a progress check every two weeks.

5. Key Performance Indicators (KPIs)

Each KPI is calculated in SQL and rebuilt as a DAX measure. If any input is missing, or if the denominator is zero, the result stays blank.

#	KPI	Formula	Unit	Business Value
1	**Net Crude Trade**	`Crude Imports - Crude Exports`	USD	Positive means net importer, negative means net exporter. Used to classify countries.
2	**Crude Import Burden**	`(Crude Imports / GDP) * 100`	% of GDP	Shows how heavy crude imports are compared with the size of the economy.
3	**Crude Export Dependence**	`(Crude Exports / GDP) * 100`	% of GDP	Shows how much a country relies on crude exports for income.
4	**Oil Intensity**	`Oil Consumption (TWh) / GDP ($ billion)`	TWh per $ billion	Shows how much oil an economy uses to produce its output.
5	**Local Currency Oil Import Bill**	`Crude Imports (USD) * FX Rate (LCU per USD)`	Local currency (LCU)	Shows the import bill in local money, so currency weakness becomes visible.
6	**Price Shock Exposure**	`Extra Import Cost = Crude Imports * shock %`; `Extra Export Revenue = Crude Exports * shock %`; `Net Exposure = shock % * Net Crude Trade`; shock % = 10%, 25%, 50%	USD and LCU	Shows how much more a country pays, or earns, if crude prices rise.

Notes on reading the KPIs

- KPI 6 assumes traded volumes stay the same. It is a simple exposure estimate, not a forecast and not a GDP-loss estimate.
- KPI 5 is a trade value in local currency. It is not the same as government spending.
- To measure currency weakness, we use `(1 - previous rate / current rate) * 100`, which shows the fall in a currency's value against the USD. This is different from the percentage rise in the local price of one dollar.
- Higher local-currency bills do not by themselves prove inflation or lower GDP. Any link to inflation is reported as a comparison, not as proof of cause.